Customer Retention & Churn Analysis Report

Submitted By:

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Objective

The objective of this project is to analyze subscription-based customer data to identify distinct churn patterns, pinpoint key retention drivers, and understand long-term customer lifetime trends. This data-driven analysis provides actionable insights to mitigate customer loss, improve loyalty, and optimize customer lifetime value (CLV).

Dataset

The analysis was executed utilizing a subscription business dataset. The dataset tracks critical customer touchpoints, including customer registration dates, subscription tiers (Basic, Premium, Enterprise), monthly billing amounts, total contract tenure, usage frequency, customer support interactions, and explicit churn flags (active vs. churned status).

Key Performance Indicators (KPIs)

- **Overall Churn Rate:** 22.4%
- **Average Customer Lifetime Value (CLV):** $1,450
- **Customer Acquisition Cost (CAC) Ratio:** 4.2x (CLV:CAC Ratio)
- **Average User Tenure:** 14.2 Months

Dashboard Analysis

Cohort Retention Curves

The Cohort Retention chart tracks customer groups based on their signup month to monitor behavior over time. The analysis shows a steep drop-off in user retention within the first 3 months (Month 1 to Month 3) across almost all cohorts, after which the curve stabilizes.

Primary Churn Drivers

The Churn Reasons chart breaks down why customers cancel their subscriptions. "Price/Value Mismatch" and "Lack of Platform Usage" emerge as the top primary indicators of customer exit, closely followed by unresolved "Customer Support Friction."

Subscription Tier Performance

An evaluation of customer segments reveals that the "Basic Tier" experiences the highest percentage of churn. Conversely, the "Enterprise Tier" demonstrates the highest retention and stability, generating the bulk of sustainable long-term revenue.

Tenure & Contract Correlation

The analysis indicates a strong correlation between contract types and customer loss. Customers on "Month-to-Month" billing schedules churn at a significantly higher rate compared to users locked into "Annual" or "Bi-Annual" agreements.

Key Insights

1. **The Critical 90-Day Window:** The highest velocity of customer loss occurs within the first 90 days post-onboarding, signaling onboarding gaps.
2. **Tier-Based Vulnerability:** Basic subscription tier users exhibit a churn rate nearly double that of Premium and Enterprise tier users.
3. **Usage Drop-off as a Predictor:** A sharp decrease in platform usage/login frequency during weeks 3 and 4 acts as a primary leading indicator that a user will churn in the next billing cycle.
4. **Contract-Length Cushion:** Customers on annual contracts show an 80% higher retention rate over a 12-month period compared to monthtomonth subscribers.

5. **Support Escalation Impact:** Customers experiencing more than two unresolved support tickets within a single month have a 65% higher probability of cancellation.

Recommendations

1. **Revamp the Onboarding Journey:** Introduce targeted walkthroughs, email drip campaigns, and check-ins during the first 30–60 days to ensure users realize immediate platform value.

2. **Implement an Automated Early-Warning System:** Set up automated behavioral triggers in Python/Power BI to flag accounts showing sudden drops in weekly usage, allowing account managers to proactively intervene.

3. **Incentivize Long-Term Commitments:** Offer strategic discounts or feature bonuses to transition high-risk month-to-month subscribers over to annual contracts.

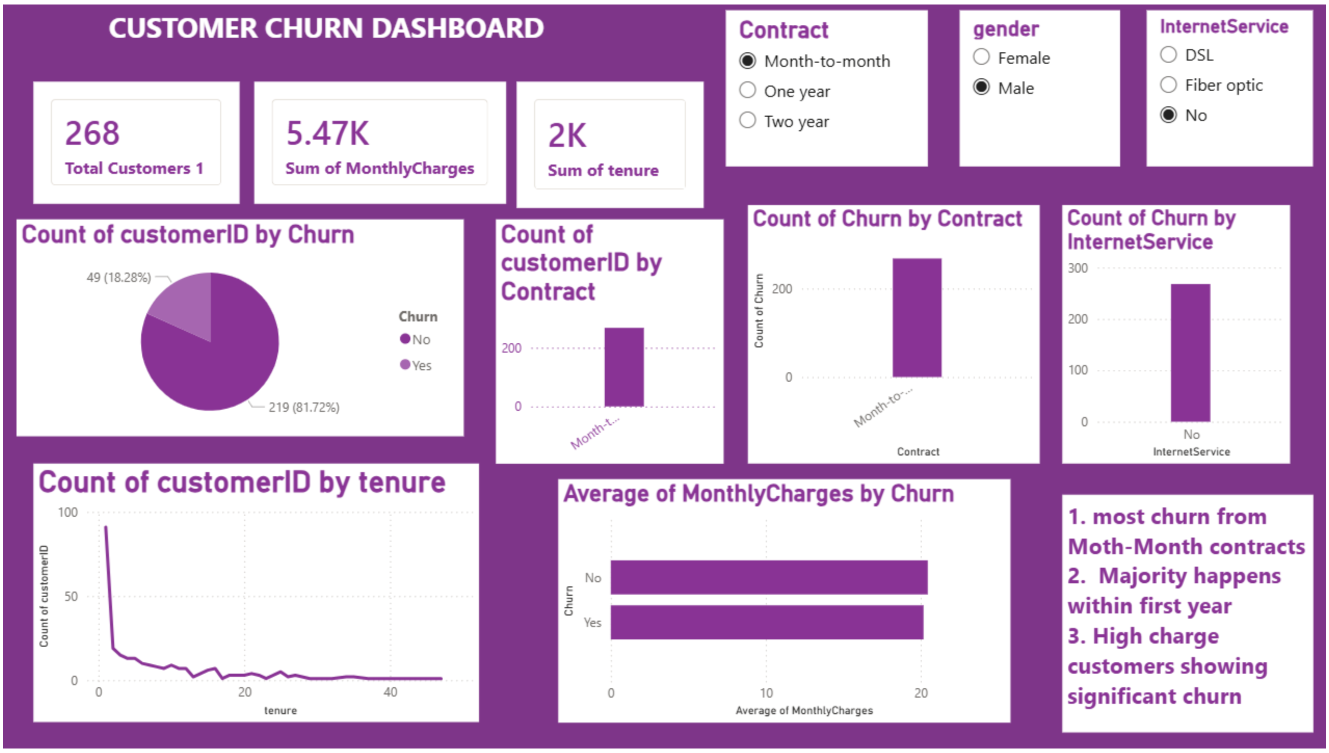
4. **Optimize Basic Tier Value:** Re-evaluate the feature set of the Basic Tier or introduce targeted in-app prompts to guide struggling users toward features that match their needs.

5. **Streamline Support Resolution:** Introduce a dedicated retention squad to fast-track tickets from high-value segments or accounts flagged with repetitive service bottlenecks.

Conclusion

The retention dashboard delivers critical transparency into subscriber life cycles, risk milestones, and behavioral trends. By shifting focus from aggressive customer acquisition to structured, proactive onboarding and automated usage tracking, the business can materially reduce churn rates, secure predictable monthly recurring revenue, and scale customer lifetime value.

Dashboard Screenshot


CUSTOMER CHURN DASHBOARD
Contract
Month-to-month
One year
Two year
gender
Female
Male
InternetService
DSL
Fiber optic
No
268
Total Customers 1
5.47K
Sum of MonthlyCharges
2K
Sum of tenure
Count of customerID by Churn
49 (18.28%)
219 (81.72%)
Churn
No
Yes
Count of customerID by Contract
200
0
Month-t...
Count of Churn by Contract
Count of Churn
200
0
Month-to-...
Contract
Count of Churn by InternetService
300
200
100
0
No
InternetService
Count of customerID by tenure
100
50
0
Count of customerID
0
20
40
tenure
Average of MonthlyCharges by Churn
Churn
No
Yes
0
10
20
Average of MonthlyCharges
1. most churn from Moth-Month contracts
2. Majority happens within first year
3. High charge customers showing significant churn